NAVYANTRA – BUILD BEYOND BOUNDARIES

24-Hour National-Level Hardware Hackathon

ABSTRACT SUBMISSION TEMPLATE

Team Name:	CORE CATALYST
Project Title:	Sparsh-Vaani: Multi-Sensory Medicine Label Reader for Differently-Abled Patients
Domain:	EMBEDDED SYSTEM

TEAM MEMBERS

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1. Problem Statement (50-75 WORDS)

People with visual, hearing, or speech disabilities often face challenges while communicating and accessing important information independently. Medicine labels can also be difficult for visually impaired users to read, which may cause confusion or incorrect use. There is a need for a simple and affordable system that can convert information between speech, text, signs, and other accessible forms of feedback.

2. Proposed Solution (100-150 WORDS)

Sparsh-Vaani is a hardware-based assistive system designed to support communication and basic medicine assistance. The system uses a camera and microphone as its main input devices. For visually impaired users, the camera captures medicine labels and OCR is used to extract information such as the medicine name, dosage, and expiry date. This information can then be provided through voice and vibration feedback. For non-speaking users, the camera recognizes a selected set of predefined hand signs and displays their corresponding meaning as text. Speech from another person can also be converted into text and shown on a display to support hearing-impaired users. Physical buttons are provided for simple control, confirmation, and repeating information. The overall aim is to provide multiple accessibility features through one affordable and easy-to-use system.

3. Methodology / Working Principle (250-300 WORDS)

The system works through three main modules: medicine assistance, sign-to-text communication, and speech-to-text communication. In the medicine assistance module, the user places a medicine strip or package in front of the camera. The Raspberry Pi captures the image, and OCR is used to identify readable information such as the medicine name, dosage, and expiry date. The extracted information is then converted into audio through a speaker, while vibration can be used to provide alerts. Buttons allow the user to repeat the information or confirm an action.

In the communication module, the camera captures hand signs made by a non-speaking user. A gesture-recognition model processes the captured image and matches it with a limited set of predefined signs. The recognized meaning is then displayed as text on an LCD or monitor. For the prototype, we will focus on selected commonly used signs rather than trying to provide complete sign-language translation.
In the speech-to-text module, a microphone captures speech from another person. Speech-recognition software converts the spoken words into text and displays them on the screen. This allows a hearing-impaired user to understand the conversation through written feedback.

Overall, the system follows a simple process: input is captured, processed using OCR, AI, or speech recognition, and then converted into an accessible output such as text, voice, vibration, or display feedback. Physical buttons are also included to make interaction easier without depending completely on a touchscreen. Each module can first be tested separately and later integrated into one common interface for the final demonstration.

4. Key Components / Technologies (100-125 WORDS)

The main hardware components include a Raspberry Pi 4 Model B as the central processing unit, an ESP32 for additional control and communication, a USB web camera for capturing images and video, a monitor for displaying information, and a keyboard and mouse for user interaction. The system can also use an ultrasonic sensor, servo motor, DC motors with chassis, L298N motor driver, buzzer, vibration or alert components, and other basic interfacing components. For communication features, speech-to-text and text-to-speech conversion will be used to support voice-based interaction. Additional components such as Bluetooth, GSM, gyroscope, stepper motors, breadboard, power supply, and supporting electronic components can be integrated depending on the selected prototype. Python and suitable AI/OCR tools will be used for processing.

5. Expected Outcome (125 WORDS)

The expected outcome is a working prototype that demonstrates how one system can provide accessible communication and medicine-related information in different forms. A visually impaired user should be able to scan a medicine package and hear the available information, while vibration can provide important alerts. A non-speaking user should be able to show selected hand signs and receive the corresponding text on the display. Speech from another person should also be converted into text so that a hearing-impaired user can read and understand the conversation. The prototype aims to reduce dependence on others for basic communication and important information. It will demonstrate the practical use of OCR, speech recognition, and simple hardware interaction in an accessibility-focused application through separate modes in one integrated prototype.

6. Innovation / Unique Feature (100-150 WORDS)

The main unique feature of Sparsh-Vaani is that it combines communication assistance and medicine information in a single system instead of focusing on only one accessibility requirement. The same platform can accept different types of inputs, such as camera images, hand signs, and speech, and convert them into suitable outputs like text, voice, and vibration according to the user's needs. The system also includes a practical healthcare application through medicine-label recognition. For the hackathon prototype, the sign-recognition module will use a limited set of predefined signs so that the demonstration remains reliable and technically realistic. Rather than trying to solve every accessibility challenge at once, the project focuses on a few useful functions that can work together in one device. The overall focus is affordability, simple interaction, and reducing communication barriers through an integrated assistive system.

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